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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Crescendo
Selected Domain	Domain 2: Orbyte Smart SME Challenge
Selected SDG Track (1–17)	8
Project / Solution Name	OpusEvents
Name (Leader)	Yaw En Qi
Email & Phone (Leader)	shaniceyaw@gmail.com, +60 162682917
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small event companies must coordinate staff, equipment, customer requests and budgets across overlapping bookings. When these records are scattered across spreadsheets and messages, managers may overlook leave, double-book staff, underestimate equipment needs, or discover staffing shortages too late. Workers face unclear schedules, while customers wait for a feasible quotation, especially during major Malaysian festive seasons such as Chinese New Year, Hari Raya, and Deepavali, when event demand surges and at the same time full-time employees take approved festive leave. Therefore, OpusEvents facilitated by Orbyte can help to tackle this problem.

OpusEvents focuses on a Malaysian SME operating around Petaling Jaya, Kuala Lumpur and Selangor. Its demonstration data contains 16 field staff and 300 historical events. These are synthetic records, not evidence of problems measured at an actual company.

The economic challenge is making timely, realistic plans without unnecessary outsourcing or misleading profit estimates. The social challenge is respecting worker availability and workload limits. Better use of reusable equipment may also reduce avoidable resource wastage. Our problem hypothesis is that one shared planning workflow, with explicit resource checks and human approval, can reduce planning effort and prevent conflicting commitments.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

OpusEvents is a working web prototype that helps an event manager turn a customer request into a reviewed resource plan. Customers can request events for up to 1,000 guests, with matching manager/customer options, start/end dates, and daily operating hours. Bookings require at least one week's notice; larger or special requests use a clearly labelled demo contact. Managers review venue, distance, resource shortages, and estimated finances, then approve and adjust crew assignments and shifts. Workers view their schedules and available open shifts. Workers can also choose to take more jobs if the schedule does not crash.

The numerical AI is a two-stage scikit-learn pipeline trained offline on RunPod: stage one predicts crew and equipment; stage two uses predicted resources and event features to estimate cost and selling price. The saved bundle runs inside the Python backend, not through a live RunPod API.

Besides that, an OpenAI-powered planning assistant answers managerial questions using a server-generated summary of the demo business. It provides advice only and cannot approve bookings. Deterministic scheduling rules, not the language model, enforce availability and resource commitments.

Confirmed bookings have shared preparation tasks, recorded expenses, remaining-budget analysis, and synthetic receipt attachments. Workers see assigned jobs in a month calendar, including overnight shift finishes, and can export to Google Calendar or ICS. Open-Meteo supplies weather/air-quality tips; Leaflet/OpenStreetMap supports venue selection.

The stack is React/TypeScript, Express, Python, PostgreSQL, and Replit App Storage. GenAI APIs, database, and cloud skills are applied; attendance remains for the team to confirm. Our prototype uses synthetic data.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Interface: React 19.1, TypeScript, Vite and Tailwind CSS. Role-specific Manager, Worker and Customer screens

API: Node.js/Express 5 with generated OpenAPI client/validators and signed demo-account cookies

Planning: Python, pandas, NumPy, scikit-learn 1.9.1 and joblib. Express talks to one persistent Python process.

Storage: PostgreSQL through psycopg; saved plans, approvals and activity survive refreshes. Replit App Storage holds private receipt files.

AI: local two-stage regression model for resources and money; optional OpenAI planning assistant via Replit AI Integrations.

External data: Open-Meteo weather/air quality; Leaflet/OpenStreetMap maps. Google Calendar links/ICS are exports, not a connected Calendar API.

Training/cloud: RunPod for offline training; Replit for web/API hosting and database.

Block diagram:

Browser → Express API → Python planning/model engine ↔ PostgreSQL.
Branches: Express → OpenAI assistant / Open-Meteo; Browser → OpenStreetMap tiles.
Offline: Synthetic training data → RunPod training → saved model → Python engine.

OpusEvents has utilized the official SDC sponsor-provided GenAI API or Cloud Platform, which are Replit, RunPod, and Orbyte.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We applied design thinking by focusing on the everyday realities of small business owners. SME staff rarely have time to learn complicated software, and owners naturally cautious about AI making financial choices. We addressed this by designing a clear, single page interface with simulated Role-Based Access Control (RBAC) and visible safety guardrails. The prototype was iterated around three roles: managers conducting planning and approval, workers needing readable assignments, and customers making consistent booking requests. It has features like colour-coded booking types, date/type organisation, multi-day booking forms, editable staffing, overnight shift handling, and a worker month calendar.

To prepare for real-world deployment in Phase 2, we identified our primary technical risk: model hallucination or GPU endpoint latency during live predictions. We built a fallback plan directly into our Replit application: if the neural network experiences delays, a deterministic math backup takes over to calculate exact inventory ratios and enforce profit margins. This ensures the app stays 100% reliable and pricing rules remain accurate under all conditions.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 8.2: "Achieve higher levels of economic productivity through diversification, technological upgrading and innovation, including through a focus on high-value added and labour-intensive sectors."
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	SME event agencies currently waste up to 40% of their operational planning time manually cross-referencing disconnected HR, inventory, and cost spreadsheets to generate a single viable event quotation.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	% Net Profit Margin Preservation per event quote and Hours Saved in Manual Resource/ Leave Reconciliation per event.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	30% reduction in manual planning time, 100% elimination of underpriced quotations, and 10 % reduction in excess rental equipment waste.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):Leaflet / React Leaflet / OpenStreetMap

Name	**Type (code/dataset/model)**	**License**
React / Express / Vite / Tailwind / UI libraries / OpenAPI tooling	Third-party code	Apache License 2.0
Python / pandas / NumPy / scikit-learn / psycopg / PostgreSQL	Code Library / Framework	Apache License 2.0/ BSD
Replit Agent	AI-assisted code/content	AI assistance
Apache License 2.0	Repository License	Apache License 2.0
Leaflet / React Leaflet / OpenStreetMap	Code and geographic data	BSD-2-Clause / Hippocratic-2.1
OpenAI API; Open-Meteo; Google Fonts; Replit identity/storage SDKs	Hosted models/APIs, data, fonts and identity/storage services	weather attribution; OFL fonts; MIT/Apache-2.0 SDK licences.

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



[24/10/2026]